

Data Mining

Hierarchical Clustering

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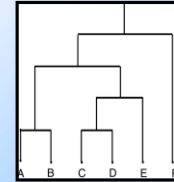
Hierarchical Algorithms

- Single Link
- Complete Link
- MST Single Link
- Average Link

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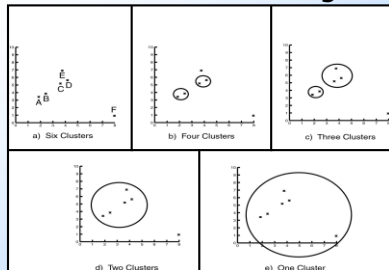
Dendrogram

- **Dendrogram:** a tree data structure which illustrates hierarchical clustering techniques.
- Each level shows clusters for that level.
 - Leaf – individual clusters
 - Root – one cluster
- A cluster at level i is the union of its children clusters at level $i+1$.



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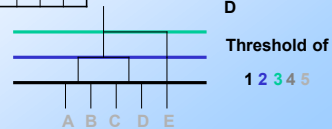
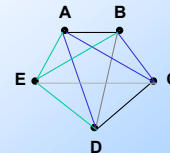
Levels of Clustering



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Agglomerative Example

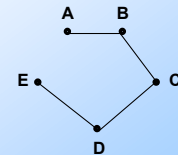
	A	B	C	D	E
A	0	1	2	2	3
B	1	0	2	4	3
C	2	2	0	1	5
D	2	4	1	0	3
E	3	3	5	3	0



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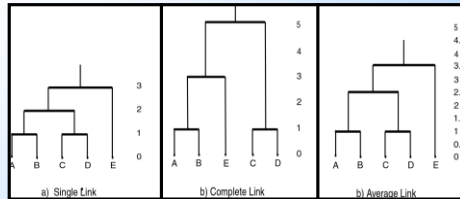
MST Example

	A	B	C	D	E
A	0	1	2	2	3
B	1	0	2	4	3
C	2	2	0	1	5
D	2	4	1	0	3
E	3	3	5	3	0



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Single Link Clustering



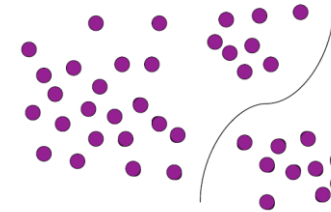
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Hierarchical Divisive: Template

1. Put all objects in one cluster
2. Repeat until all clusters are singletons
 - a) choose a cluster to split
 - what criterion?
 - b) replace the chosen cluster with the sub-clusters
 - split into how many?
 - how split?
 - “reversing” agglomerative => split in two
- cutting operation: cut-based measures seem to be a natural choice.
 - focus on similarity across cut - lost similarity
- not necessary to use a cut-based measure

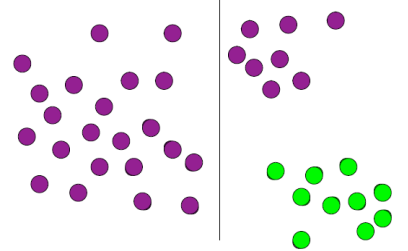
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An Example: 1st cut



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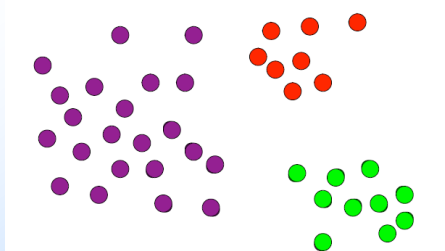
An Example: 2nd cut



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An Example: stop at 3 clusters



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Divisive (top bottom) procedures

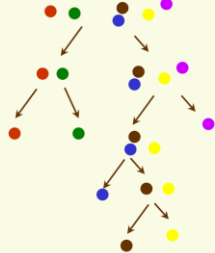
- Start with all samples in one cluster
- Form hierarchy by splitting the “worst” clusters

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Divisive Hierarchical Clustering

- Any "flat" algorithm which produces a fixed number of clusters can be used
 - set $c = 2$



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Challenges & Solutions

- It is **difficult** to select merge or split points
- No **backtracking**
- Hierarchical clustering **does not scale** well: examines a good number of objects before any decision of split or merge
- One promising directions to solve these problems is to combine hierarchical clustering with other clustering techniques: **multiple-phase clustering**

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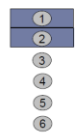
3.3.3 BIRCH

- BIRCH**: Balanced Iterative Reducing and Clustering Using Hierarchies
- Agglomerative** Clustering designed for clustering a **large** amount of numerical **data**
 - It overcomes the two difficulties in agglomerative clustering methods: (1) scalability and (2) the inability to undo what was done in the previous step.
- What Birch algorithm tries to solve?
 - Most of the existing algorithms DO NOT consider the case that **datasets** can be **too large** to fit in main **memory**
 - They DO NOT concentrate on **minimizing** the number of **scans** of the dataset
 - I/O **costs** are very high
- The complexity of BIRCH is $O(n)$ where n is the number of objects to be clustered.

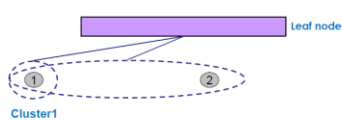
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BIRCH: The Idea by example

Data Objects



Clustering Process (build a tree)

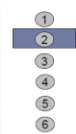


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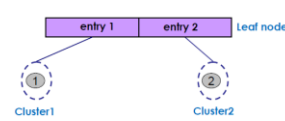
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BIRCH: The Idea by example

Data Objects



Clustering Process (build a tree)



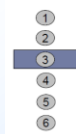
Leaf node with two entries

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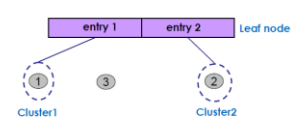
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BIRCH: The Idea by example

Data Objects



Clustering Process (build a tree)

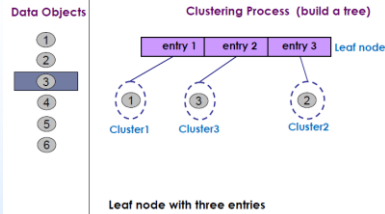


entry1 is the closest to object 3

If cluster 1 becomes too large by adding object 3, then split the cluster

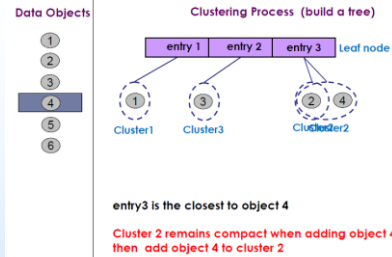
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BIRCH: The Idea by example



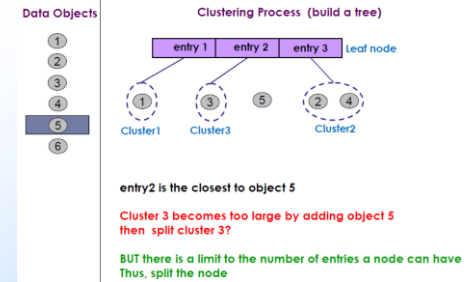
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BIRCH: The Idea by example



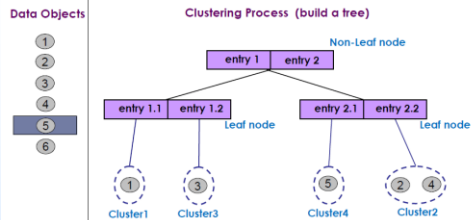
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BIRCH: The Idea by example



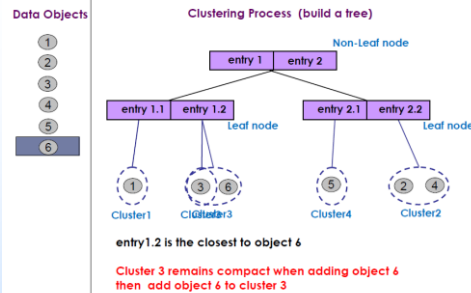
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BIRCH: The Idea by example



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BIRCH: The Idea by example



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BIRCH: Key Components

Clustering Feature (CF)

- Summary of the statistics for a given cluster: the 0-th, 1st and 2nd moments of the cluster from the statistical point of view
- Used to: compute centroids, and measure the compactness and distance of clusters

CF-Tree

- height-balanced tree
- two parameters:
 - number of entries in each node
 - The diameter of all entries in a leaf node
- Leaf nodes are connected via *prev* and *next* pointers

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Clustering Feature

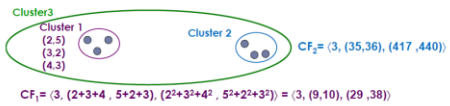
Clustering Feature (CF): $CF = (N, LS, SS)$

N : Number of data points

LS : linear sum of N points: $\sum_{i=1}^N X_i$

SS : square sum of N points: $\sum_{i=1}^N X_i^2$

$CF_3 = CF_1 + CF_2 = (3+3, (9+35, 10+36), (2^2+4^2, 3^2+4^2)) = (6, (44, 46), (446, 478))$



$CF_1 = (3, (2+3+4, 5+2+3), (2^2+3^2+4^2, 5^2+2^2+3^2)) = (3, (9, 10), (29, 38))$

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Properties of Clustering Feature

- CF entry is a **summary** of statistics of the cluster
- A **representation** of the cluster
- A CF entry has **sufficient information** to calculate the centroid, radius, diameter and many other distance measures
- Additively** theorem allows us to **merge** sub-clusters incrementally

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